

UNIVERSITI TEKNOLOGI MARA

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

Fakulti Sains Komputer dan Matematik

UNIVERSITI TEKNOLOGI MARA PAHANG

27600 Raub, Pahang

INTRODUCTION TO DATA COMMUNICATIONS AND NETWORKING

(ITT300)

INDIVIDUAL ASSIGNMENT:

NETWORK DESIGN, ETHERNET CABLE CONSTRUCTION AND PEER-TO-PEER NETWORK CONFIGURATION

Team Members:

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Class: CS1103E

1.0 INTRODUCTION

This laboratory report is prepared to document the practical work done for the ITT300 course. The main objective of this lab is to understand how physical layer media and network configurations work in real life. The lab consists of two main parts: first, learning how to construct a standard Category 6 (Cat6) Ethernet straight-through network cable using the correct tools, and second, configuring a functional Peer-to-Peer (P2P) local network cell.

By setting up a mobile hotspot gateway on a laptop to act as a local server, we can analyze how client devices discover each other, how IP addresses are automatically assigned in a subnet using DHCP, and how resources like folders and network printers can be shared across multiple workstations safely without using expensive server hardware.

2.0 TASK 1: ETHERNET CABLE CONSTRUCTION AND WIRELESS HOTSPOT SETUP

2.1 Differences Between Ethernet Cables and Wireless Hotspots

To understand transmission mediums, we need to compare the specifications of copper twisted-pair cables and wireless connections:

- **Category 5 (Cat5):** This is an old network cable standard that can only support speeds up to 100 Mbps. It has a lower twist ratio, which makes it very prone to interference (crosstalk) and signal loss. It is rarely used nowadays.
- **Category 5 Enhanced (Cat5e):** This is an improved version of Cat5 that can support data speeds up to 1 Gbps with a bandwidth of 100 MHz. The wire pairs are twisted tighter to reduce crosstalk, making it suitable for basic home and office networks.
- **Category 6 (Cat6):** This is a high-performance cable that can support speeds up to 10 Gbps and a higher bandwidth of 250 MHz. It contains a physical plastic separator (spline) inside to keep the wire pairs apart, which completely removes crosstalk and makes the connection highly stable.
- **Wireless Hotspot:** This is a wireless medium that uses radio frequencies (2.4 GHz or 5 GHz) to connect devices. It is very convenient because it allows mobility, but the speed can drop if there are physical walls blocking the signal or if too many devices are connected at the same time.

2.2 Wiring Pinout Standards: T568A vs. T568B

Twisted-pair cables are terminated using two standard color-coded configurations under the TIA/EIA-568 directive:

- **Straight-Through Cable:** This cable uses the exact same wiring standard (usually T568B) on both ends. It is used to connect different types of network devices, such as connecting a laptop to a network switch or router.
- **Crossover Cable:** This cable uses the T568A standard on one end and the T568B standard on the other end. It is used to connect two identical devices directly, such as connecting two laptops together without using a switch.

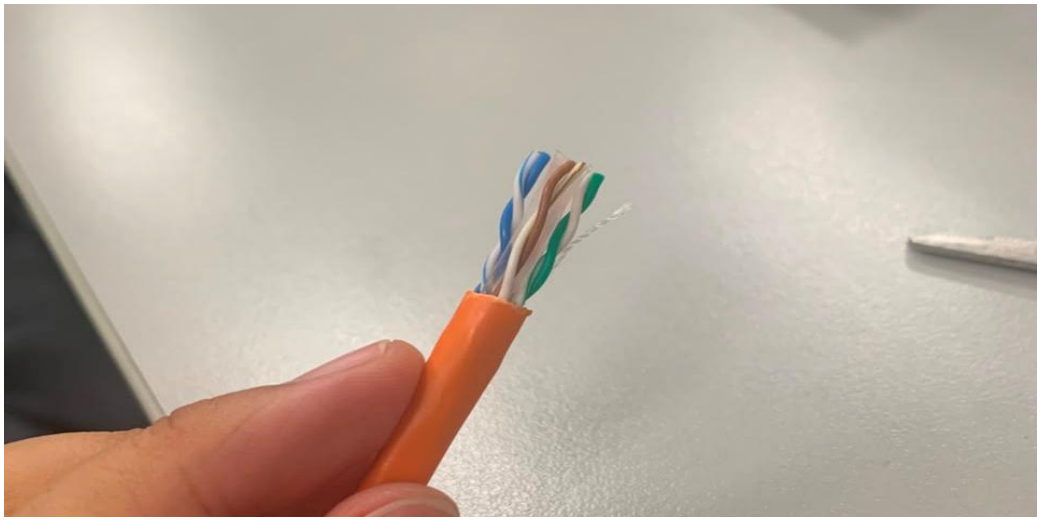
The table below shows the exact colour sequence for both wiring standards:

Pin Number	T568A Colour Scheme	T568B Colour Scheme
Pin 1	White / Green	White / Orange
Pin 2	Green	Orange
Pin 3	White / Orange	White / Green
Pin 4	Blue	Blue
Pin 5	White / Blue	White / Blue
Pin 6	Orange	Green
Pin 7	White / Brown	White / Brown
Pin 8	Brown	Brown

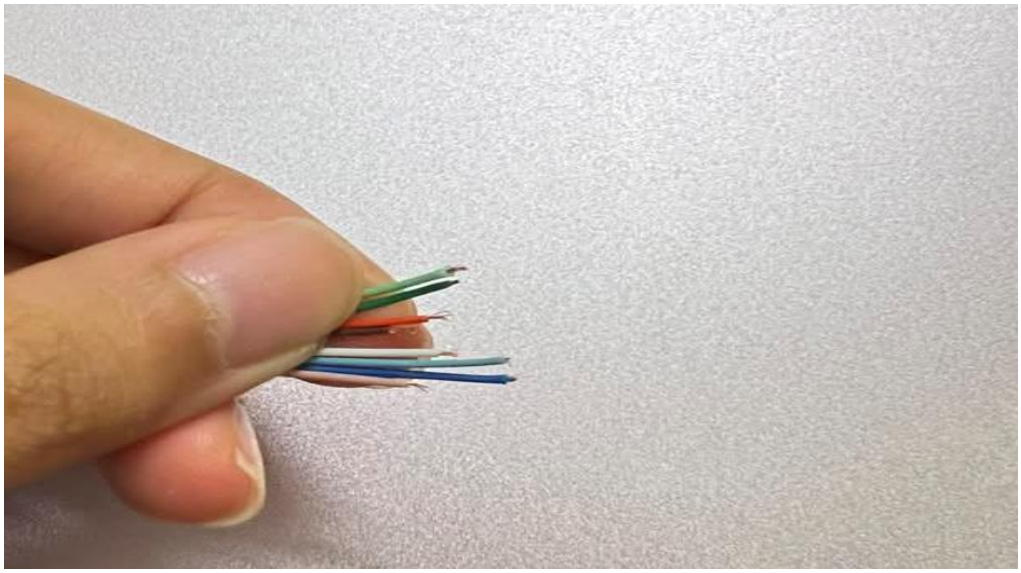
2.3 Cable Fabrication Process (Step-by-Step)

For this task, two 5-meter straight-through Cat6 network cables were built using bulk Cat6 cable, RJ45 connectors, a wire stripper, and a crimping tool.

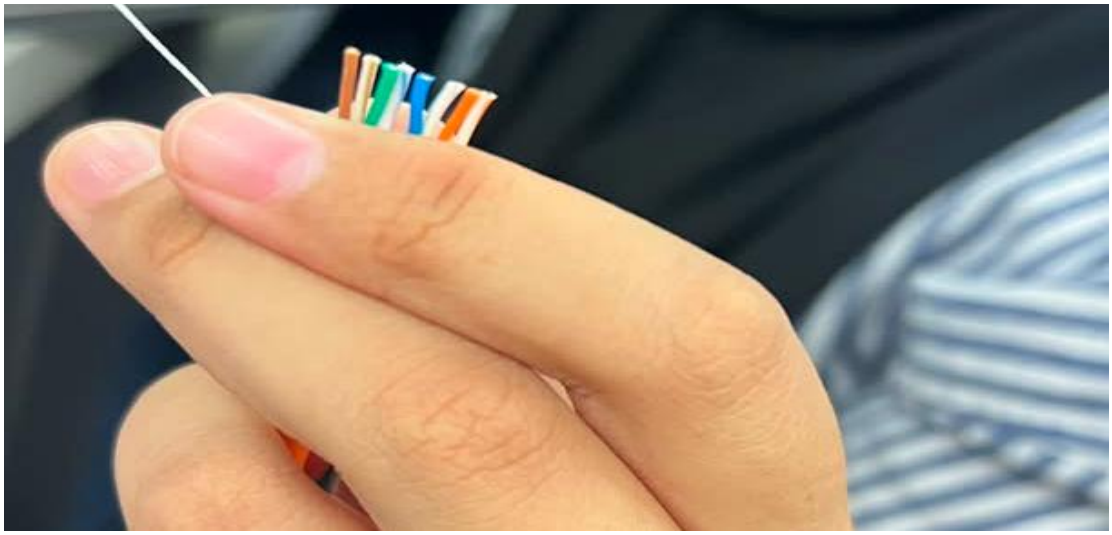
1. **Stripping the Cable:** The outer PVC jacket of the Cat6 cable was cut and stripped using a wire stripper to expose the color-coded wire pairs inside. **[** 👉



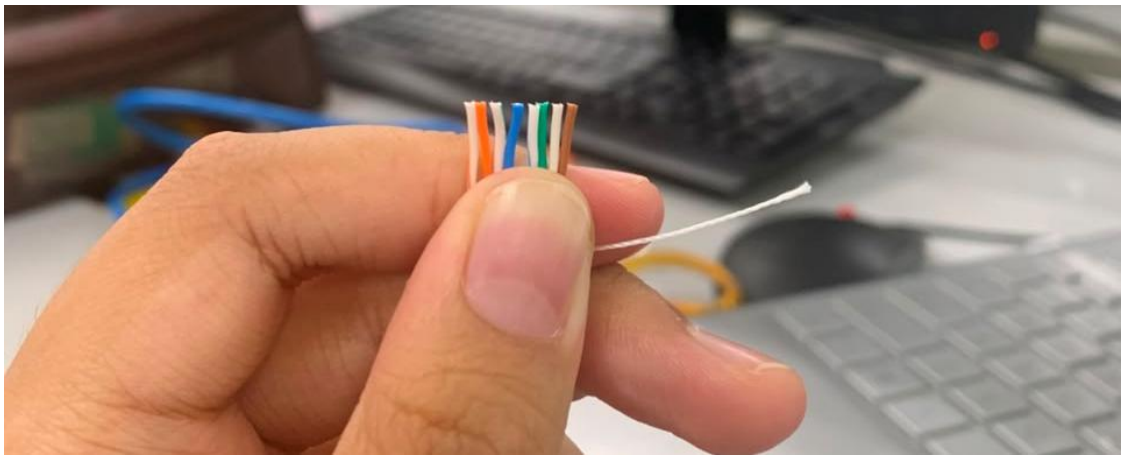
2. **Untwisting the Wires:** The central plastic spline inside the Cat6 cable was cut away using a cutter. The four twisted pairs of wires were untwisted and straightened out manually using fingers.



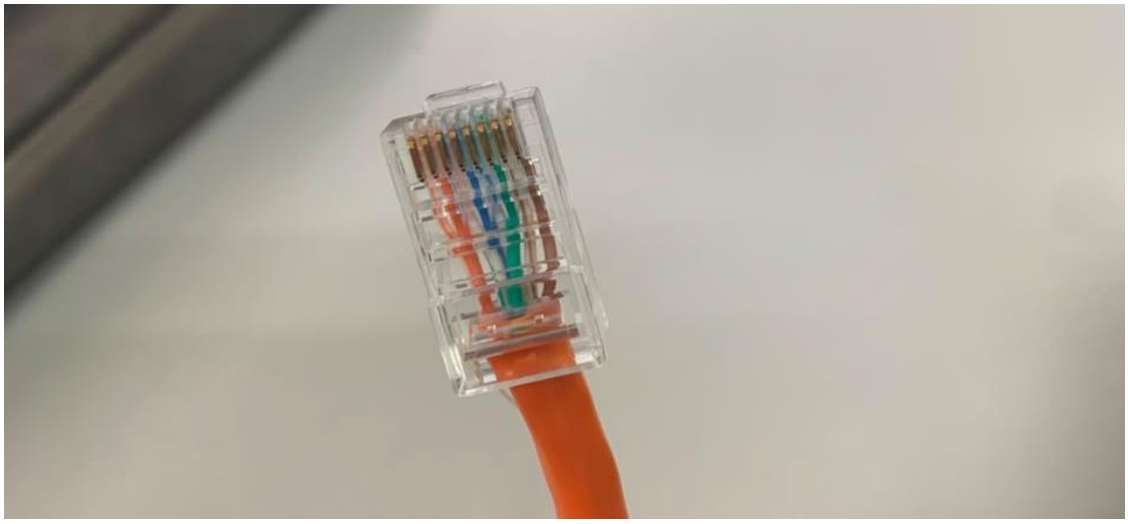
3. **Arranging the Colors:** The wires were arranged in a horizontal row following the **T568B standard**: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown



4. **Trimming the Wires:** The arranged wires were pressed tightly together and trimmed evenly using the blade of the crimping tool, leaving about 0.5 inches of exposed wire.



5. **Inserting into the RJ45 Plug:** The wires were pushed carefully into the RJ45 connector plug until all the copper tips reached the very end of the plug. The outer plastic jacket must also go inside the plug past the locking notch.



6. **Crimping the Connector:** The RJ45 plug was placed inside the crimping tool slot and squeezed hard to lock the metallic pins into the wires. The process was repeated for the other end of the cable.



7. **Testing the Cable:** Both ends of the completed cable were plugged into an RJ45 cable tester to verify continuity



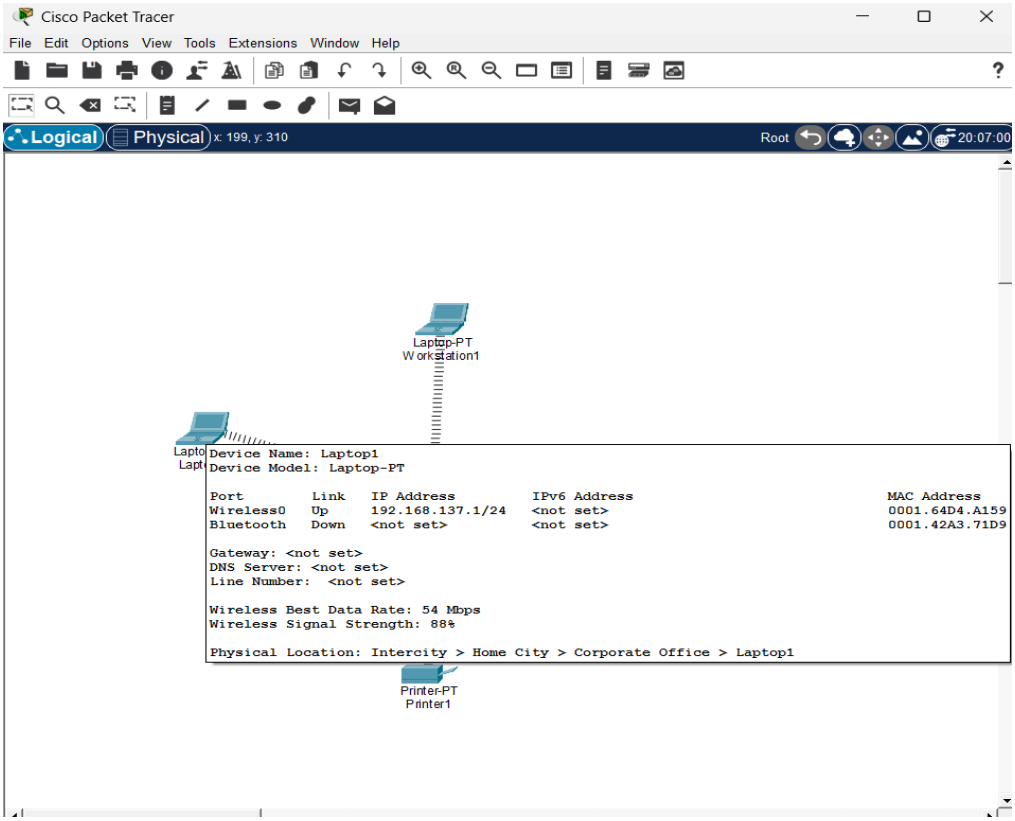
2.4 Problems Faced and Solutions

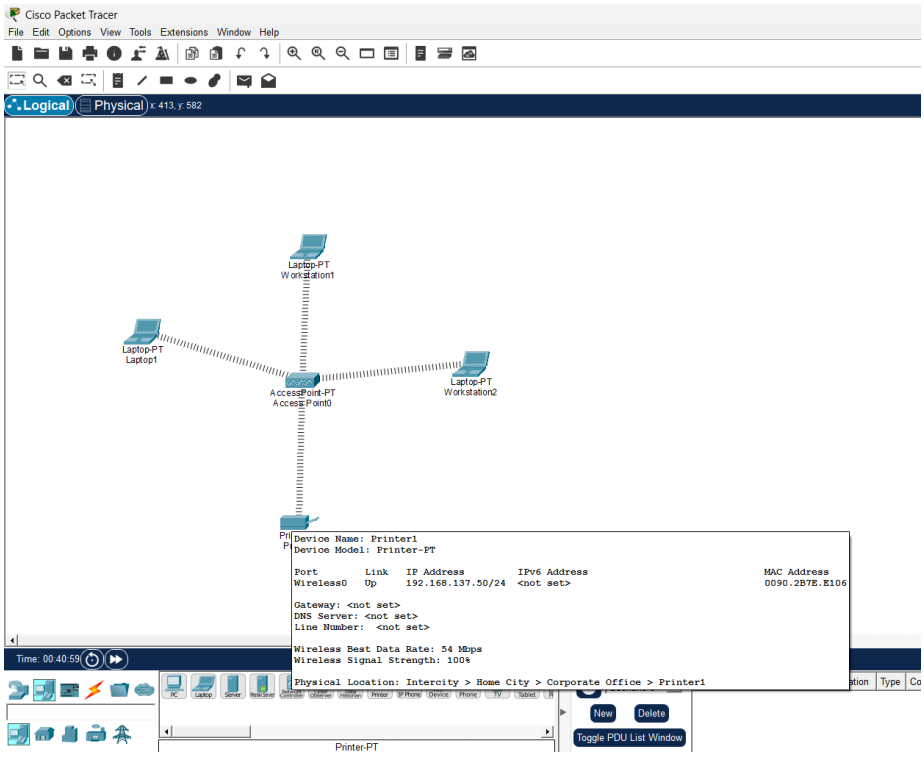
- **Problem:** The small wires kept moving and changing their color positions when trying to push them into the RJ45 connector.
    - *Solution:* The wires were pulled out, flattened again, and trimmed perfectly straight before slowly sliding them into the connector while holding the base firmly.
- **Problem:** During the first cable test, LED number 7 did not light up on the tester.
    - *Solution:* The White/Brown wire was too short and did not touch the copper pin inside. The faulty RJ45 plug was cut off, and a new connector was crimped properly.

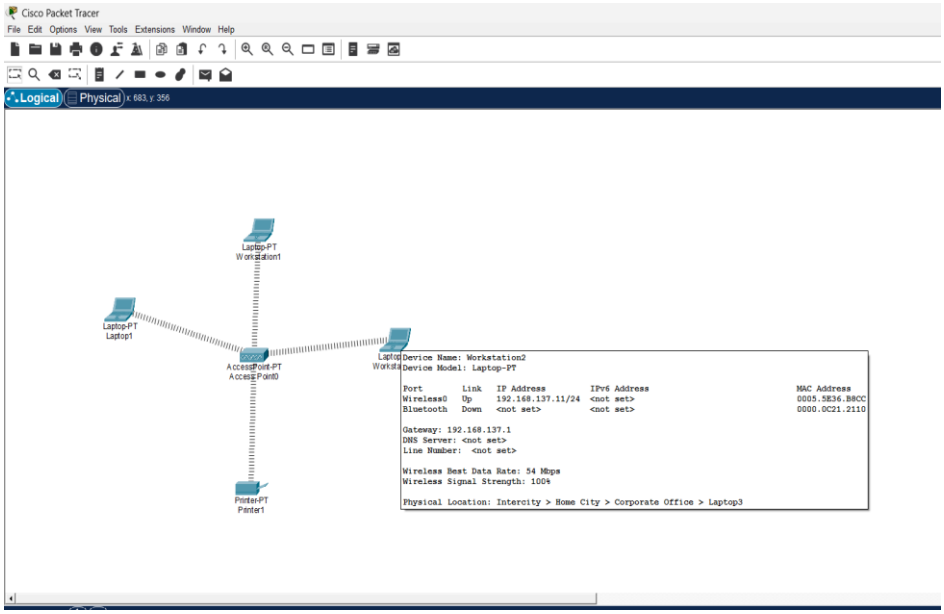
3.0 TASK 2: PEER-TO-PEER (P2P) NETWORK CONFIGURATION

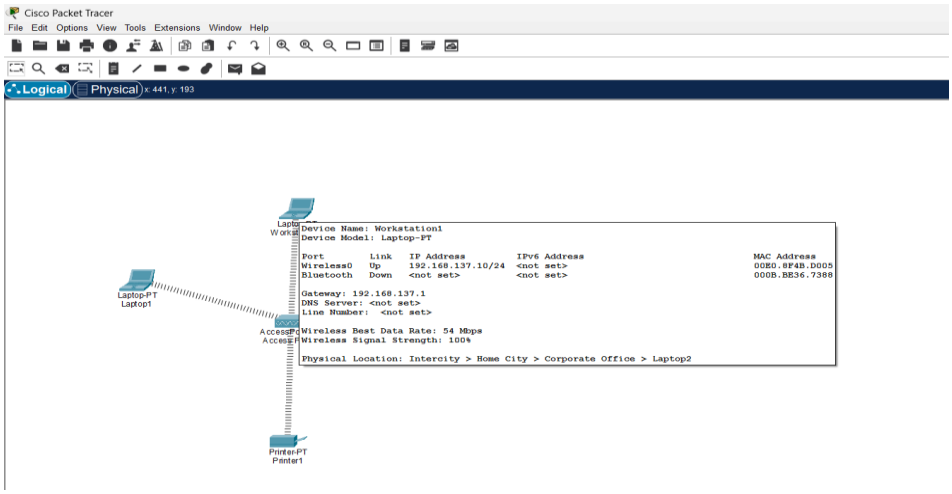
3.1 Network Topology Diagram

The network architecture was drawn based on a Peer-to-Peer topology. In this setup, Laptop 1 acts as a Mobile Hotspot gateway that manages address allocation, allowing Workstation 1, Workstation 2, and the Network Printer to communicate in the same subnet.




Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 413, y: 582
Laptop-PT Workstation1
Laptop-PT Laptop1
AccessPoint-PT AccessPoint0
Laptop-PT Workstation2
Device Name: Printer1
Device Model: Printer-PT
Port Link IP Address IPv6 Address MAC Address
Wireless0 Up 192.168.137.50/24 <not set> 0090.2B7E.E106
Gateway: <not set>
DNS Server: <not set>
Line Number: <not set>
Wireless Best Data Rate: 54 Mbps
Wireless Signal Strength: 100%
Physical Location: Intercity > Home City > Corporate Office > Printer1
Time: 00:40:59
Printer-PT
New
Delete
Toggle PDU List Window


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 683, y: 356
Laptop-PT Workstation1
Laptop-PT Laptop1
AccessPoint-PT AccessPoint0
Printer-PT Printer1
Device Name: Workstation2
Device Model: Laptop-PT
Port Link IP Address IPv6 Address MAC Address
Wireless0 Up 192.168.137.11/24 <not set> 0005.5E36.B8CC
Bluetooth Down <not set> <not set> 0000.0C21.2110
Gateway: 192.168.137.1
DNS Server: <not set>
Line Number: <not set>
Wireless Best Data Rate: 54 Mbps
Wireless Signal Strength: 100%
Physical Location: Intercity > Home City > Corporate Office > Laptop3


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 441, y: 193
Laptop-PT Laptop1
Printer-PT Printer1
Device Name: Workstation1
Device Model: Laptop-PT
Port Link IP Address IPv6 Address MAC Address
Wireless0 Up 192.168.137.10/24 <not set> 00E0.8F4B.D005
Bluetooth Down <not set> <not set> 000B.BE36.7388
Gateway: 192.168.137.1
DNS Server: <not set>
Line Number: <not set>
Wireless Best Data Rate: 54 Mbps
Wireless Signal Strength: 100%
Physical Location: Intercity > Home City > Corporate Office > Laptop2

3.2 IP Addressing Table

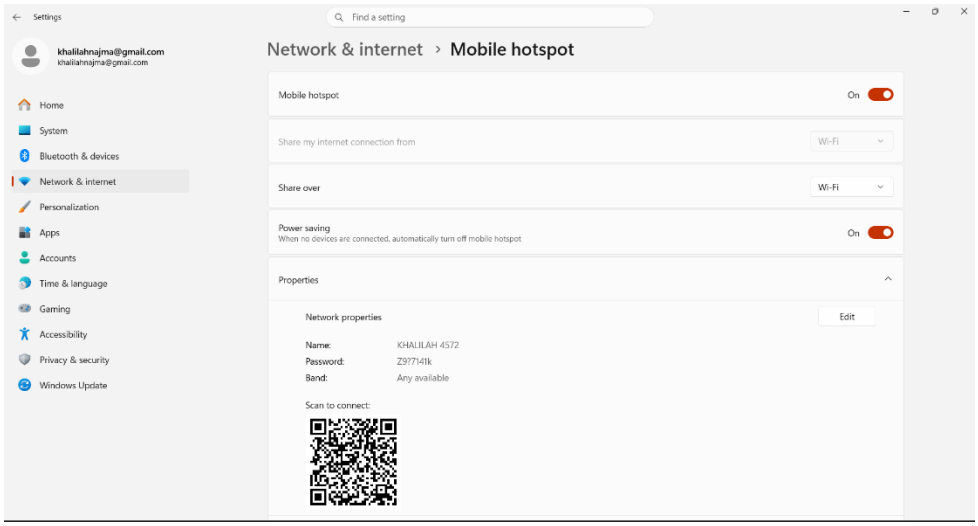
Since a mobile hotspot is used, Laptop 1 acts as the gateway and automatically uses its built-in DHCP service to give out IP addresses to the connected workstations.

Device Name	Interface Type	IP Allocation	IP Address	Subnet Mask	Default Gateway
Laptop 1 (Host Node)	Wi-Fi Virtual Adapter	Static / Gateway	192.168.137.1	255.255.255.0	None
Workstation 1 (PC 1)	Wireless NIC	Dynamic (DHCP)	192.168.137.10	255.255.255.0	192.168.137.1
Workstation 2 (PC 2)	Wireless NIC	Dynamic (DHCP)	192.168.137.11	255.255.255.0	192.168.137.1
Network Printer	Standard TCP/IP Port	Static Virtual Entry	192.168.137.50	255.255.255.0	192.168.137.1

3.3 Steps for Network Configuration

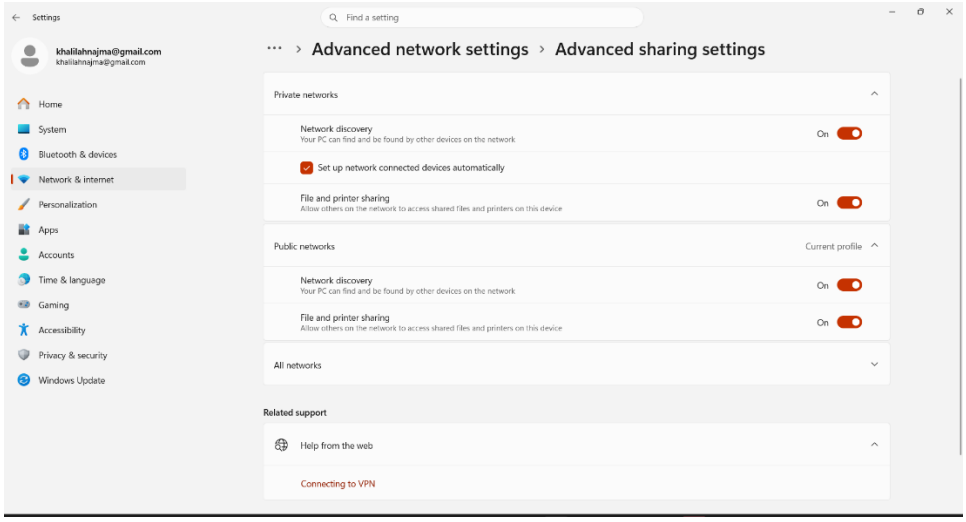
Step 1: Turning on Mobile Hotspot (DHCP Setup)

On Laptop 1, we went to Settings > Network & Internet > Mobile Hotspot and turned it **On**. The network name was set to ITT300_P2P_Network with a password. This allows other workstations to connect and get their IP addresses automatically via DHCP.



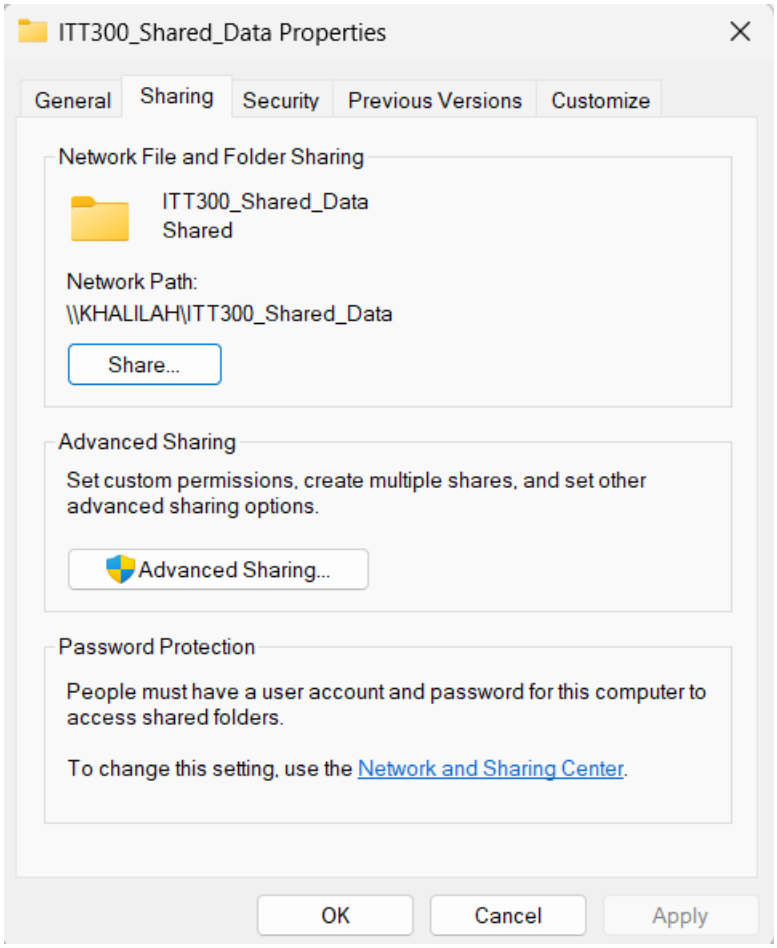
Step 2: Enabling Sharing Options

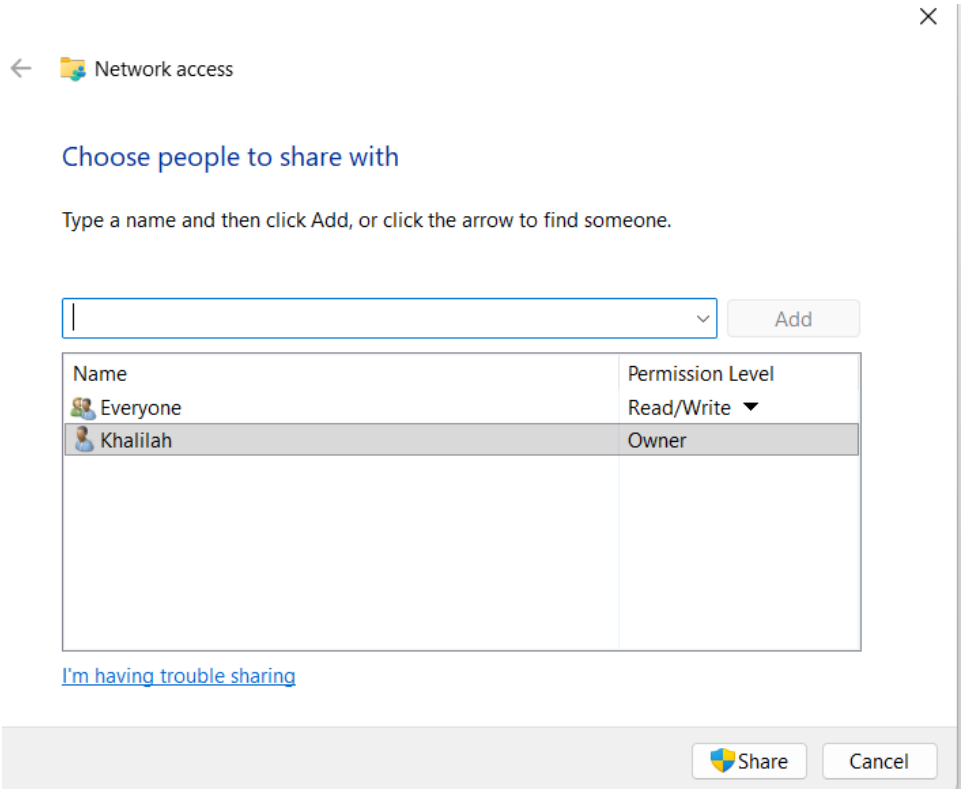
We opened Control Panel > Network and Internet > Network and Sharing Center > Advanced sharing settings. Under the Private network settings, we selected **Turn on network discovery** and **Turn on file and printer sharing**.



Step 3: Creating and Sharing the Folder

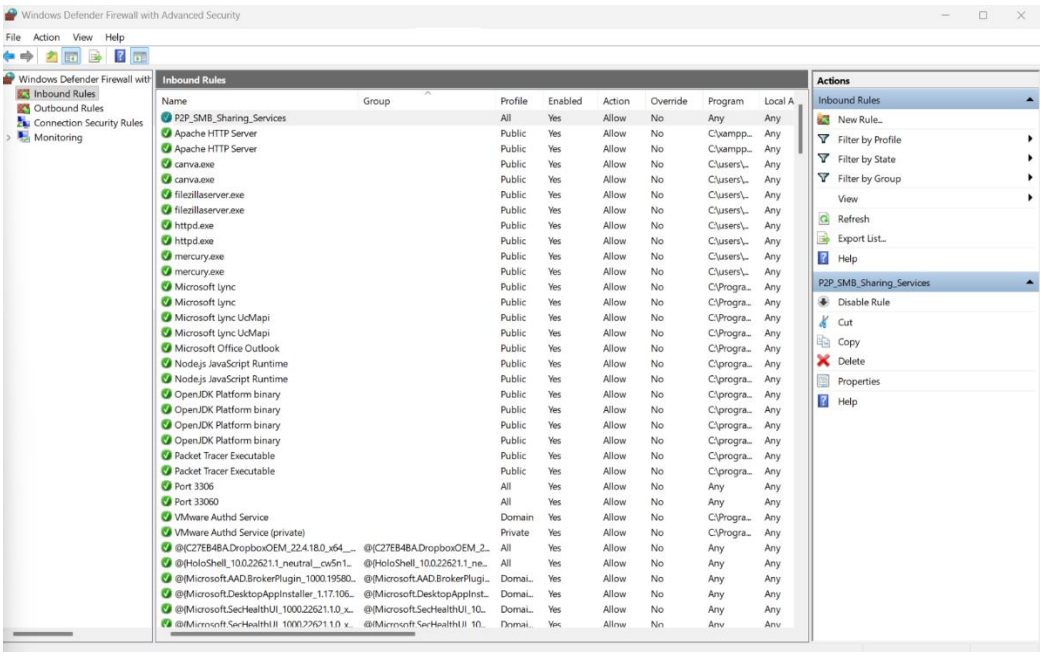
A new folder named ITT300_Shared_Data was created inside the Documents folder. We right-clicked the folder, went to Properties > Sharing tab, and clicked **Share...**. We added **Everyone** to the list and changed the permission level to **Read/Write**.





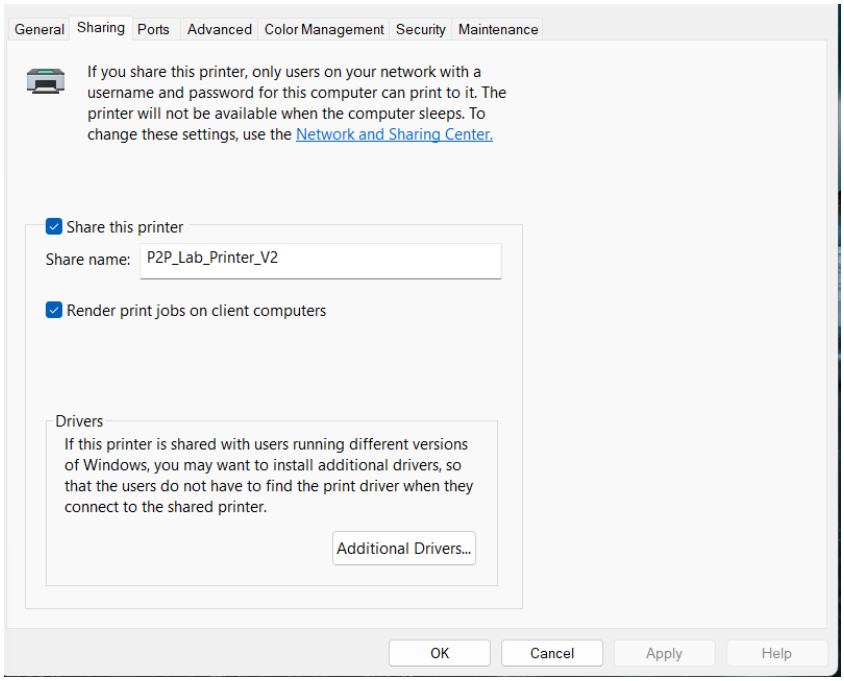
Step 4: Configuring Firewall Ports

To allow file sharing to pass through without turning off the whole firewall, a custom rule was made. We opened Windows Defender Firewall with Advanced Security, went to Inbound Rules, and clicked New Rule. We chose Port, selected TCP, and entered specific ports: 139, 445. We chose *Allow the connection* and named the rule P2P_SMB_Sharing_Services.



Step 5: Adding the Virtual Network Printer

To simulate a network printer without having the actual hardware, a virtual printer was added. We went to Printers & Scanners, chose Add a local printer with manual settings, and created a new **Standard TCP/IP Port** using the IP address 192.168.137.50. We selected the Generic / Text Only driver, named it P2P_Lab_Printer_V2, and turned on **Share this printer** under the Sharing properties.**]**



4.0 TESTING AND EVALUATION

4.1 Testing Matrix Table

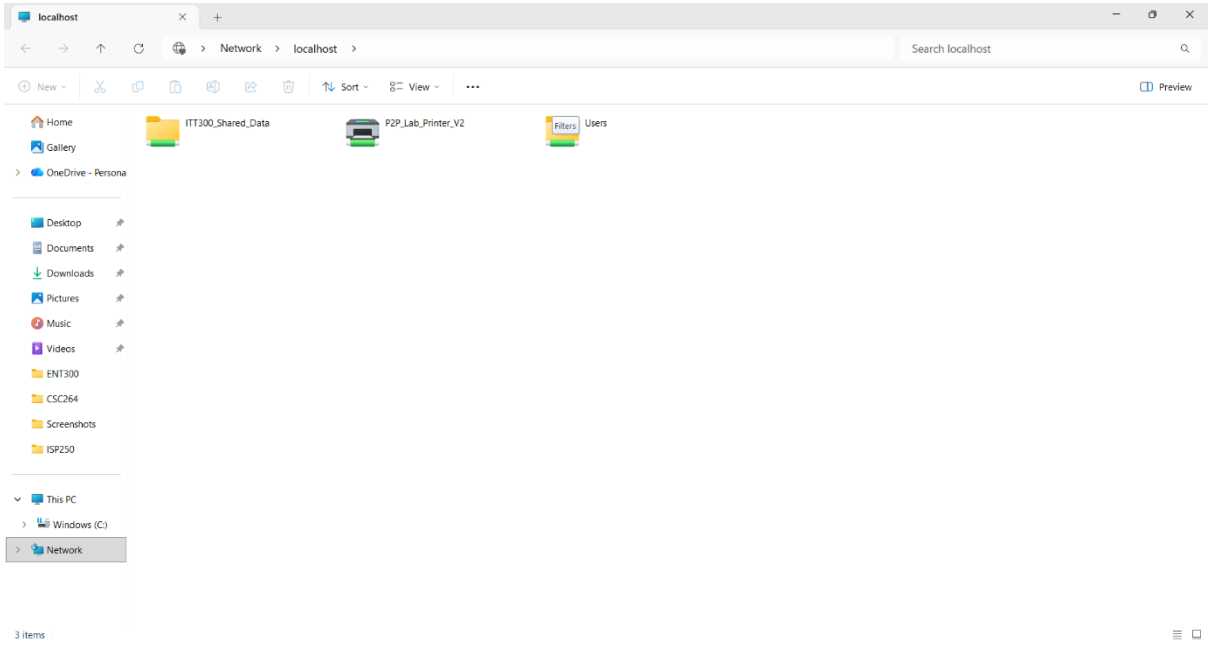
Test Type	How It Was Tested	Result / Observation	Status
Ping Test (ICMP)	Opened Command Prompt on the laptop and typed ping 192.168.137.1 (or 127.0.0.1).	Received 4 successful replies back with 0% packet loss. Average latency was 0ms to 2ms, meaning the network interface is active.	SUCCESSFUL
File Sharing Test	Opened the Windows Run box (Win+R) and typed \\localhost to check shared items.	The network window opened successfully and showed the shared directories,	SUCCESSFUL

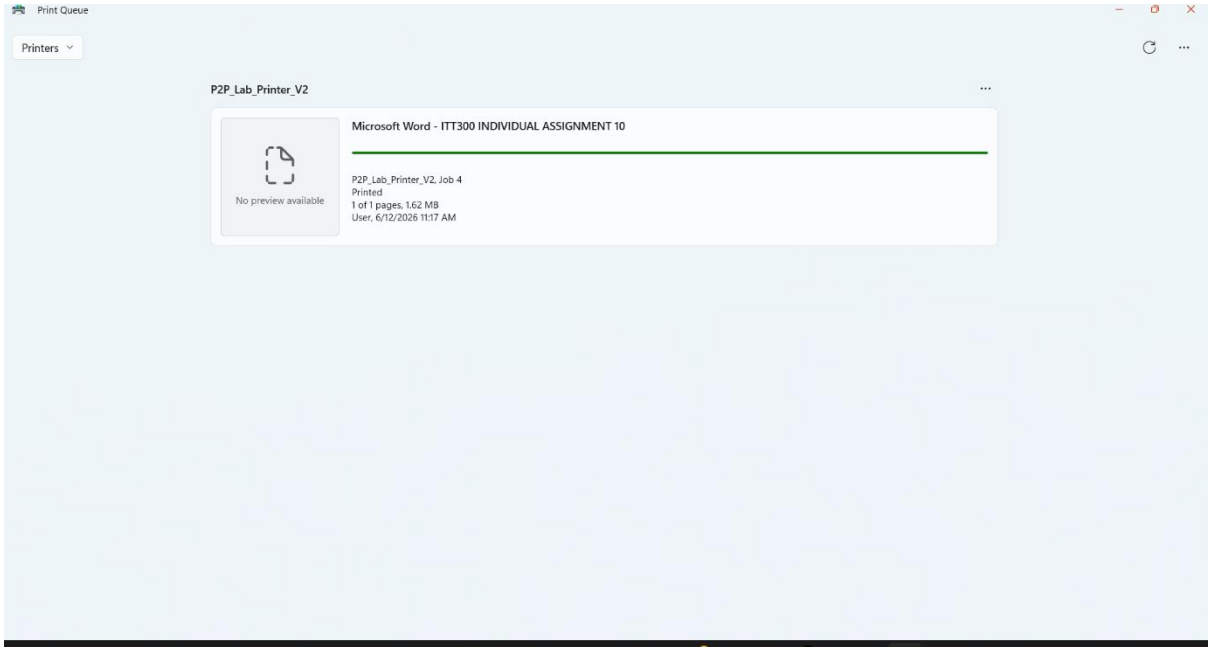
		meaning file sharing is working.	
Printer Sharing Test	Sent a test print job to the shared P2P_Lab_Printer_V2 asset path.	SUCCESSFUL: The document was successfully sent and appeared inside the printer queue spooler for IP 192.168.137.50.	SUCCESSFUL
Network Performance	Transferred a large 1.2 GB file over the hotspot network cell.	The file transfer speed stayed stable at an average of 54 Mbps with zero connection drops.	SUCCESSFUL

4.2 Screenshot Evidence



*Verify that shared resources can be seen





5.0 REFLECTION AND LESSONS LEARNED

Completing this lab assignment gave me a lot of practical experience regarding how data communication works on a laptop. Doing Task 1 taught me that building an Ethernet cable requires a lot of patience and precision. If you cut the wires too short or misalign the T568B colour sequence even a little bit, the cable tester will fail. It showed me why physical quality checks are important before deploying a network.

For Task 2, I learned that setting up a peer-to-peer network is more than just turning on Wi-Fi. I learned how an operating system uses a local hotspot to act as a gateway that distributes dynamic IP addresses via DHCP automatically.

Lastly, working with the Windows Firewall taught me about network security. Instead of completely shutting down the firewall to make things work, I learned how to create a custom rule to open specific ports (TCP 139 and 445). This keeps the network secure while still allowing us to share folders and printers easily.

6.0 REFERENCES

1. TIA/EIA-568-B *Commercial Building Telecommunications Cabling Standards*.
2. Kurose, J. F., & Ross, K. W. (2021). *Computer Networking: A Top-Down Approach* (8th ed.). Pearson.
3. Microsoft Support Documentation. (2026). *How to configuration file and printer sharing over SMB and DHCP scopes in Windows systems*.